

LAB 3 — WRITTEN JUSTIFICATION

Digital Campus Library Reservation Gateway

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Architecture selected: Layered Architecture

I selected Layered Architecture because our library system has three main types of work: user interaction, library operations and data storage. Keeping these separate makes the system easier to understand.

Reason 1: The different responsibilities are separated. The UI handles the interaction, while the Reservation & Holding Queue component handles the FIFO queue, holds and cancellations. The database only stores the required information.

Reason 2: It is easier to make changes later. For example, changing the database structure should not require changing the complete user interface.

Security: UC-03 is for the Head Librarian, so access control is important. The Authentication & Access Control component can check whether the person is allowed to perform queue administration.

Performance: The search part is separated from the other operations. This makes it easier to optimize catalogue search and pagination for the large number of books mentioned in the requirements.

Main components

Component	Role
Member / Librarian UI	Student and librarian interaction
Authentication & Access Control	Login and authorization
Catalog & Search	ISBN and catalogue search
Reservation & Holding Queue	Holds, FIFO queue and queue changes
Notification Service	Member notifications
Fine Management	Overdue fine calculation
Library Database	Stores system data

Interfaces: UI–Authentication, UI–Catalog, UI–Reservation/Queue, Reservation/Queue–Notification, Reservation/Queue–Database and Fine Management–Database.